Rank-nine fusion rings of multiplicity two

Theoretical obstructions, coherent constructions and exact certification

FusionFrontier research prepared for Sébastien Palcoux

AI-generated research; independent human verification and formalization pending

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Abstract

We study the 2,083 supplied rank-nine based fusion rings of exact multiplicity two. The current certified partition consists of 2,033 general complex exclusions, 32 rings with unitary models and 18 unresolved rings. The proofs combine formal-codegree and subring obstructions, signed sphericalization, central induction, modular trace constraints, Morita theory and explicit categorical constructions. A Galois restriction theorem simultaneously excludes six rings by half-integral restriction multiplicities. A forced central $\mathrm{Rep}(S_3)$ gives a further structural exclusion. Rank-one module categories and the dimension-twenty Hopf classification settle three additional rings, while explicit relative cocycles construct three integral models. Every decision is bound to an original tensor, a proof dependency and exact replay data. We classify existence of a categorification of a based ring, without classifying all its associators.

Verification scope. These are AI-generated mathematical claims. Separate mathematical audits and exact computational replays are provided; independent human verification and Lean formalization are pending.

Contents

1 The census, identifiers and accepted partition **2**

2 Uniform criteria and finite certificates **3**
2.1 Formal codegrees, cyclotomicity and subrings 3
2.2 Central induction with representation degrees 3
2.3 Signed reduction, neutral components and Morita theory 4

3 A Galois restriction theorem **4**

4 A forced central symmetric category excludes 516 **5**

5 The dimension-forty rank-one extension family **6**
5.1 A fiber functor gives two exclusions 6
5.2 Explicit Hopf data and a unitary model 7

6 Three relative-cocycle models **8**
6.1 A construction with its twisted character formula 8
6.2 The model of dimension 120 8
6.3 Two models on the same group and subgroup 8

7 The half-E_6 family and the model inventory **9**
7.1 Products and associative zesting 9
7.2 A central-fermion shear with two coherent actions 9
7.3 All 32 positive tensors 10

8 What remains unresolved **11**

9 Exact evidence and reproduction **11**

1 The census, identifiers and accepted partition

A fusion ring has a distinguished basis $1 = X_0, X_1, \ldots, X_8$, nonnegative integer constants $X_iX_j = \sum_k N_{ij}^k X_k$, an involution $i \mapsto i^*$, and the usual unit and Frobenius reciprocity identities. Multiplicity means $\max_{i,j,k} N_{ij}^k$. The source file through multiplicity two contains 2,225 tensors: 142 of multiplicity one and 2,083 of exact multiplicity two. Its complete producer run finished all five duality strata. Enumeration completeness is producer-reported; tensor validity and exact uniqueness are separately checked. We do not repeat the entire enumeration here.

The label $R9_2_n$ denotes the nth *exact-multiplicity-two* tensor in source order, not necessarily physical line n of the mixed source file. Every ledger entry also supplies its physical line and SHA-256 of compact JSON, without a trailing newline. The 142 multiplicity-one labels retain explicit basis transports to the multiplicity-one classification; fourteen of these transports are nonidentity. No literal unchanged-basis assertion is needed. The original tensor, rather than a dimension profile, determines case identity.

Theorem 1.1 (Certified partition of the supplied exact-M2 census)**.** *The accepted ledger partitions the 2,083 source rings into 2,033 rings excluded from fusion categorification over $\mathbb{C}$, 32 admitting a unitary fusion categorification, and 18 unresolved rings. Every exclusion has a scope-valid theoretical argument and an exact certificate or an explicit transfer from an excluded tensor. Every realization has categorical existence data and a complete equality of its 9^3 structure constants with the source tensor under the recorded based permutation.*

Proof. The finite partition is the exact union of the source-bound exclusion and model records described below. The verifier checks all 2,083 tensor hashes, all 568 explicit negative-subring transfers and all 23,328 coefficients of the 32 positive tensors. It rejects any overlap between exclusion and realization records. Each family-specific proof supplies the mathematical implication used by its finite receipt. The model and exclusion arguments are audited separately from these bookkeeping checks. Complete data and replay commands appear in Section 9. □

Subring-proof discipline. A closed based subring transfers an exclusion only when that subring has an actual negative proof. Absence from a positive model catalog has no logical force. In particular the rank-three half-E_6 ring

$$f^2 = 1, \qquad fx = xf = x, \qquad x^2 = 1 + f + 2x$$

has a unitary categorification [5, 6]. It is never used as a negative subring. The audit withdraws 26 claims based on that omission, constructs models for seven affected ambient rings, and replaces the proofs for $R9_2_97, 98, 1489$ by their different $\mathrm{NG}(C_2^2, 2)$ subrings. For those subrings the established near-group restriction $0 < \kappa < |G| - 1$ gives the exclusion. The corrected code contains an explicit negative registry with tensor identities and proof dependencies.

2 Uniform criteria and finite certificates

2.1 Formal codegrees, cyclotomicity and subrings

For an irreducible representation E of a based algebra let $e_E = \dim E$ and f_E be its formal codegree. The fusion Casimir acts by $e_E f_E$; the degree factor must be retained in a noncommutative algebra. Formal codegrees of a fusion category are d-numbers. If the minimal polynomial of an algebraic integer a is $t^n + a_1 t^{n-1} + \cdots + a_n$, the exact divisibilities $a_n^i \mid a_i^n$ express the d-number test. Failure supplies a finite integer witness. Character cyclotomicity is used with its proper scalar character hypotheses; a nonabelian Galois-group witness cannot be replaced by an approximate eigenvalue calculation [1, 2].

For an integral ring a hypothetical categorification is pseudo-unitary. Distinguished central dimensions D/f_E restrict to sums of integral source dimensions, hence are integers. In the noncommutative calculation the central Casimir eigenvalue is divided by e_E before imposing this condition. Near-group and Tambara–Yamagami restrictions use their stated group and coefficient hypotheses. Tensor closure, dual closure and a full based permutation are checked before any proper-subring transfer.

The baseline dependency audit reviews all 588 original structural uses. After removing the 26 invalid half-E_6 uses, the other 562 have direct proofs or explicit lower negative bindings. The 454 rank-six through rank-eight uses match excluded lower tensors exactly; 166 stored direct witnesses replay; 24 small-rank uses are supplied with fresh direct certificates. These counts describe overlapping audit classes, not a sum of distinct excluded ambient rings.

2.2 Central induction with representation degrees

Let $\mathcal{C}$ be pseudo-unitary, $D = \mathrm{FPdim}(\mathcal{C})$ and $d_i = \mathrm{FPdim}(X_i)$. Write $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ for forgetting the half-braiding and I for its right adjoint. The distinguished simple $A_E \subset I(1)$ has

$$[I(1) : A_E] = e_E, \qquad \dim A_E = D/f_E.$$

For every central simple Z, its restriction column $v_Z = ([F(Z) : X_i])_i$ is nonnegative integral, and

$$[I(X_i) : Z] = v_{Z,i}, \qquad \dim Z = \sum_i v_{Z,i} d_i.$$

The full noncommutative induction identity is

$$FI(X) = \bigoplus_{Y \in \mathrm{Irr}(\mathcal{C})} Y \otimes X \otimes Y^*.$$

Consequently the columns satisfy the exact Gram identity

$$\sum_Z v_{Z,i} v_{Z,j} = [FI(X_i) : X_j].$$

In a commutative ring this is the usual Casimir Gram matrix; the simplification is not used when the multiplication is noncommutative.

Every summand of $I(1)$ has twist one. The first and second twist traces in $I(X_i)$, and the traces of cyclic rotations in tensor powers, give further necessary equalities. Arithmetic and Galois conditions on central dimensions and the integrality of modular character values restrict the finite column domains. Retaining a larger domain only weakens an exclusion test.

Proposition 2.1 (Exact separating certificate)**.** *Suppose a finite rational system $Au = b$, $u \geq 0$, is proved necessary for a categorification. If an integer vector y satisfies $A^t y \geq 0$ and $b^t y < 0$, there is no such categorification.*

Proof. A solution would give $b^t y = u^t A^t y \geq 0$, a contradiction. Every entry of the matrix and ray is checked exactly, after clearing denominators. A failed numerical optimization or an unsuccessful search for y has no negative mathematical implication. □

2.3 Signed reduction, neutral components and Morita theory

A criterion formulated for pseudo-unitary categories is transferred to arbitrary complex categories only through an established existence reduction. Weak integrality supplies pseudo-unitarity directly. In the indicated irrational cases, exact signed-sphericalization and character certificates show that a hypothetical categorification has a pseudo-unitary Galois conjugate of the same based ring. This is an equivalence of existence questions, not an assertion that every original category is pseudo-unitary. The signed audit checks the complete finite sign-character domains and nonzero-product constraints. A signed gate alone never excludes a ring.

The graded and Morita arguments use actual invertible module categories. For matrix conventions where source indices act on rows, module matrices represent the opposite multiplication: $M_i M_j = \sum_k N_{ji}^k M_k$. The linking-category Gram argument is applied with this convention, including noncommutative cases. A feasible Gram relaxation does not construct an invertible bimodule.

The accepted finite families, their whole-domain reductions and exact rays are retained in the replay archive. The 36 intermediate exclusions are grouped in Table 1. All their proofs and scope reductions passed a separate audit. The folded SO(3) argument for $R9_2_920$ includes the odd-boundary step: split a boundary strand by the normalized trivalent vertex and retract by the bigon identity. Thus spanning and relations extend from even boundaries without an unsupported bending assertion.

Table 1: The 36 family exclusions. Listed integers are suffixes of $R9_2_n$.

Family	Cases
Signed nonzero products	686,690,726,728,736,1187,1241,1377,1424
Joint integral center	301,352,366,385,725,727,1028,1134,1238,1448,1465,1501, 1712,1778,1989
Prime descent	907
Solvable rank-two module	1071,1724
Joint quadratic center	1569,1886
Folded fermionic SO(3)	920
Solvable central invertibles	1356,1933
Quadratic neutral block	500,1235
Fixed-block Morita obstruction	396,1265

3 A Galois restriction theorem

The following criterion reconstructs an entire central restriction from the source algebra. It does not enumerate the rest of the center.

Theorem 3.1 (Galois-unit restriction)**.** *Let R be a commutative fusion ring whose FP dimensions generate a real quadratic field K. Suppose every formal codegree belongs to K, and write $\sigma \neq 1$ for its automorphism. Let C_R be the regular Casimir matrix. Its distinct irreducible minimal factors have degree one or two. Define $p \in \mathbb{Q}[t]$ modulo their product by*

$$p(t) \equiv 1 \pmod{t-a}, \qquad p(t) \equiv t^2/n \pmod{t^2 - at + n}.$$

If R has a pseudo-unitary categorification, the unit row of $p(C_R)$ is a nonnegative integer vector with unit entry one.

Proof. Let χ_h run through all scalar irreducible characters, with formal codegree f_h, and let $A_h \subset I(1)$ be their distinguished central simples. Put $D = \mathrm{FPdim}(\mathcal{C})$ and let B correspond to

$\sigma \circ \mathrm{FPdim}$. Then $\dim A_h = D/f_h$, $\dim B = D/\sigma(D)$, and $[F(B) : 1] = 1$. The central-character identity and symmetry of the unnormalized modular S-matrix [3] give

$$\begin{aligned}(D/f_h)\chi_h(FB) &= S_{A_h,B} = S_{B,A_h} \\ &= \frac{D}{\sigma(D)}(\sigma \circ \mathrm{FPdim})(FA_h) = \frac{D}{\sigma(D)}\sigma(D/f_h) = \frac{D}{\sigma(f_h)}.\end{aligned}$$

Restriction multiplicities are integers, which justifies commuting σ with the dimension of FA_h. Hence

$$\chi_h(FB) = f_h/\sigma(f_h).$$

For rational f_h this is one; for a quadratic root with norm n it is f_h^2/n. Rational Chinese remaindering therefore constructs the displayed p. All characters agree on FB and $p(C_R)$, and the semisimple commutative regular representation is faithful. Thus $p(C_R)$ is exactly multiplication by FB, proving its required integrality and unit entry. The characters themselves need not take their values in K; only their formal codegrees do. □

Corollary 3.2. *The rings* R9_2_114, 128, 1487, 1635, 1636, 1959 *have no complex fusion categorification.*

Proof. Their FP field is $\mathbb{Q}(\sqrt{3})$, with $D = 96 + 48\sqrt{3}$. The forced restrictions, in source order, are

114	$(1, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, 1, 0, 0)$
128	$(1, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0)$
1487	$(1, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$
1635	$(1, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 1)$
1636	$(1, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, \frac{1}{2}, \frac{1}{2})$
1959	$(1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0)$.

Each contains nonintegral multiplicities. The exact independent replay constructs the polynomial by a rational remainder linear system, checks its values in every character component, verifies the full multiplication matrix identity, and checks dimension $D/\sigma(D) = 7 + 4\sqrt{3}$. The separately audited signed gates reduce every general complex existence question here to pseudo-unitary existence. The contradiction therefore has general scope. □

Remark 3.3 (Duality variant and noncommutative extension). A dual-invariant scalar character labels a self-dual distinguished central simple: $S_{A^*,Z} = S_{A,Z^*}$ and the character formula identify their S-rows. In cases 1487 and 1959 a character of codegree 24 forces dimension $4 + 2\sqrt{3}$. Its restriction must contain the unit and exactly one of a pair of dual simples of dimension $2 + 2\sqrt{3}$, which is impossible. For a noncommutative irreducible E of degree e_E, the central eigenvalue is χ_E/e_E. The same proof gives $f_E/\sigma(f_E)$, while the Casimir eigenvalue is $e_E f_E$. The integer e_E cancels in the Galois ratio. The extension yields integral nonnegative vectors for 1848 and 1975 and makes no exclusion for them. All eleven applicable commutative quadratic positive controls pass the criterion.

4 A forced central symmetric category excludes 516

Proposition 4.1. *The ring* R9_2_516 *has no fusion categorification over* $\mathbb{C}$.

Proof. Use its audited general-to-pseudo-unitary gate. The exact codegree data force precisely one distinguished object of dimension one and one, A, of dimension two in $I(1)$; every other such dimension exceeds four. Its restriction is $F(A) = 1 + g$, with $g^2 = 1$. Uniqueness makes A self-dual. The unit multiplicity of $F(A)^2$ is two. Since all distinguished multiplicities in $I(1)$ are

one, A^2 has exactly one other distinguished summand in addition to the unit. The dimension bound forces that summand to be A, leaving an invertible z with $F(z) = g$. Thus

$$A^2 = 1 + z + A, \qquad zA = A, \qquad z^2 = 1.$$

The verified character of A has value one at g. The character formula and balancing give $\theta_z = 1$; all three objects have trivial double braiding on their closed fusion subcategory. Positive dimensions identify this symmetric category as Tannakian $\mathrm{Rep}(S_3)$.

Condense the central regular algebra $1 + z$. Its image $1 + g$ is connected, so its module category $\mathcal{E}$ is a fusion category. The spherical trace descends after division by two, giving its positive spherical structure. Free and fixed source orbits under g give simple dimensions

$$1^3, \quad d^3, \quad e^3, \qquad d = 1 + \sqrt{3}, \quad e = 2 + \sqrt{3}.$$

The local $1 + z$-modules in the central $\mathrm{Rep}(S_3)$ form $\mathrm{Rep}(C_3)$. Under center descent [4, Theorem 3.20], its two nontrivial invertibles forget separately to the unit of $\mathcal{E}$: their sum is the free module on $F(A) = 1 + g$, namely two units. Distinct central structures on the unit give a faithful C_3 of tensor automorphisms of the identity. Their eigencharacters define a faithful dual C_3 grading, whose components have equal FP dimension [1]. Equating the rational and $\sqrt{3}$ coefficients, a component contains either three dimension-d simples or one each of dimensions $1, d, e$. The neutral component contains a unit, so it has dimensions $1, d, e$.

Its rank-three fusion rules are forced by dimension and duality:

$$X^2 = 1 + X + Y, \quad XY = X + 2Y, \quad Y^2 = 1 + 2X + 2Y.$$

Its three distinguished restriction columns are $(1, 0, 0), (1, 1, 1), (1, 2, 2)$, with dimensions $1, 4 + 2\sqrt{3}, 7 + 4\sqrt{3}$. Their twist-one contribution in $I(X)$ is $M = 18 + 10\sqrt{3}$. The total dimension of $I(X)$ is $T = 30 + 18\sqrt{3}$. Its first twist trace is zero, since $X \neq 1$. All other twists have modulus one, requiring $M \leq T - M$. But $2M - T = 6 + 2\sqrt{3} > 0$, the contradiction. An independently reconstructed labelled-S ray against 14,128 columns supplies an additional exact certificate, without being needed for this structural proof. □

5 The dimension-forty rank-one extension family

5.1 A fiber functor gives two exclusions

Proposition 5.1. *The rings R9_2_1225 and R9_2_1673 have no complex fusion categorification.*

Proof. In either ring the eight labels other than 5 form a fusion subring $\mathcal{D}$ of dimensions $1^4, 2^4$ and FP dimension 20. The other simple m satisfies

$$Xm = mX = d_X m, \qquad m^2 = \sum_X d_X X.$$

The additive category generated by m is a rank-one left $\mathcal{D}$-module category, equivalent to Vec. Its action gives a strong exact faithful tensor functor $\mathcal{D} \to \mathrm{End}(\mathrm{Vec}) = \mathrm{Vec}$. Ordinary Tannaka reconstruction gives $\mathcal{D} \simeq \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension 20. This uses neither a pivotal structure nor Hopf self-duality.

The complete dimension-twenty semisimple list consists of group algebras, their duals, and the two exceptional Gelaki algebras A_0, A_1 [7, 8, 9]. Function Hopf algebras are pointed; group representation rings are commutative. The two exceptional character rings are also commutative, as the direct calculation below proves. Both source neutral rings are noncommutative, witnessed respectively by the products of labels $(1, 2)$ and $(1, 3)$. Thus none of the possibilities can occur. □

5.2 Explicit Hopf data and a unitary model

For an odd prime p and $t = 0, 1$, write $B(a, e, b) = x^a \delta_e g^b$, with $a \in \mathbb{Z}/p$ and $e, b \in \mathbb{Z}/2$. The Andruskiewitsch–Natale construction [9] has

$$\begin{aligned} B(a,e,b)B(c,f,d) &= \delta_{e,f}(-1)^{tebd} B(a+(-1)^b c, e, b+d), \\ \Delta B(a,e,b) &= \sum_{s=0,1} B(a,s,b) \otimes B((-1)^s a, e-s, b), \\ \epsilon B(a,e,b) &= \delta_{e,0}. \end{aligned}$$

Its four one-dimensional simples $L(e, s)$ have $x = 1$ and $g = (-1)^s i^{te}$. The remaining $p-1$ simples $V(e, j)$, $1 \le j \le (p-1)/2$, have dimension two and x-eigenvalues $\zeta_p^{\pm j}$. Their characters are invariant under $a \mapsto -a$, so inversion in the coproduct is invisible in every character convolution. This proves commutativity of both exceptional character rings. Explicitly,

$$\begin{aligned} L(e,s)L(f,r) &= L(e+f, s+r+tef), \\ L(e,s)V(f,j) &= V(e+f, j), \\ V(e,j)V(f,k) &= V(e+f, j+k) + V(e+f, j-k), \end{aligned}$$

where $V(e, -j) = V(e, j)$ and $V(e, 0) = L(e, 0) + L(e, 1)$.

Put $\zeta = \zeta_p$. An explicit symmetric nondegenerate ordinary Hopf pairing and its inverse copairing are

$$\begin{aligned} P(B(a,e,b), B(c,f,d)) &= \delta_{e,d}\delta_{f,b}\zeta^{ac} i^{tbd}, \\ \gamma &= \frac{1}{p} \sum_{a,c,e,b} \zeta^{-ac} i^{-teb} B(a,e,b) \otimes B(c,b,e). \end{aligned}$$

The positive Haar functional is $h(B(a, e, b)) = \delta_{a,0}\delta_{b,0}/2$. Set $\lambda = \sqrt{4p}\, h$. Directly,

$$(\Delta \otimes 1)\gamma = \gamma_{13}\gamma_{23}, \qquad (1 \otimes \Delta)\gamma = \gamma_{12}\gamma_{13},$$

$$(S \otimes 1)\gamma = (1 \otimes S)\gamma, \qquad (\lambda \otimes \lambda)(1 \otimes S)\gamma = 1.$$

The copairing is symmetric, $S^2 = 1$, and the distinguished grouplike is one. These are precisely the conditions of [10, Corollary 3.19]; rigidity is supplied by their Section 3.6. Hence there is a coherent fusion category $\mathrm{Rep}(A_t) + \mathrm{Vec}$ with the displayed odd rules. In their dual-Hopf convention the self-duality map is $S\Gamma_0$, but their formula for the associator inserts S^{-1}. Thus the actual associator copairing is the γ displayed above, without another antipode.

The compact involution is

$$B(a,e,b)^* = (-1)^{teb} B(-(-1)^b a, e, b).$$

The Haar inner product makes these basis vectors orthogonal of squared norm $1/2$. Each block of γ is a Fourier bicharacter element times a unitary and a phase. The odd Fourier associator is

$$\phi(B(a,e,b)) = \frac{i^{-teb}}{\sqrt{p}} \sum_u \zeta^{-(-1)^e ua} B(u,b,e),$$

a permutation of normalized unitary Fourier blocks. The remaining nonidentity associators are representations of γ, the Hopf transport maps $h \otimes v \mapsto h_{(1)} \otimes h_{(2)} v$, and their inverses. Haar invariance makes these transports unitary. In the odd-even-odd case right multiplication composed with the antipode is a star representation, with the corresponding opposite algebra understood. All associators are therefore unitary.

For $p = 5$, $t = 1$ gives $R9_2_965$, with source-to-model map $(0, 2, 3, 6, 7, 8, 5, 4, 1)$. The $t = 0$ construction recovers $R9_2_21$ as a positive control. The independent checker verifies both copairing identities, normalization, all character convolutions, and all 729 coefficients.

6 Three relative-cocycle models

6.1 A construction with its twisted character formula

Let G be a finite group, $H < G$, $\omega \in Z^3(G, U(1))$, and let ψ be normalized with $\delta\psi = \omega^{-1}|_H$ in the convention

$$\psi(a,b)\psi(ab,c) = \omega(a,b,c)\psi(b,c)\psi(a,bc).$$

The subgroup algebra $A = \bigoplus_{h\in H} h$ in Vec_G^ω is connected and separable. Its bimodules form the unitary group-theoretical category $\mathcal{C}(G,\omega,H,\psi)$ [1, 11]. For a double-coset representative u put $K_u = H \cap uHu^{-1}$. On its degree-u fiber, the stabilizer representation has multiplier

$$\alpha_u(a,c) = \frac{\psi(a,c)\omega(a,u,u^{-1}cu)}{\omega(a,c,u)\omega(u,u^{-1}au,u^{-1}cu)\psi(u^{-1}au,u^{-1}cu)}.$$

This follows by writing the left and right module actions:

$$L_aL_c = \psi(a,c)\omega(a,c,v)^{-1}L_{ac}, \quad R_cR_a = \omega(v,a,c)\psi(a,c)R_{ac}, \quad R_cL_a = \omega(a,v,c)L_aR_c.$$

One defines $\rho_u(a) = R^{-1}_{u^{-1}au}L_a$ and substitutes these identities. For the relative tensor product the balancing relation is

$$R_h(m_u)\otimes n_v = \omega(u,h,v)m_u \otimes L_h(n_v).$$

In a stabilizer trace at degree uv, a fixed coset contributes

$$\frac{\omega(u,h,v)}{\omega(a,u,v)\omega(u,v,b)}\chi_u^M(a)\chi_v^N(h), \quad h = u^{-1}au, \quad b = (uv)^{-1}a(uv).$$

Projective character inner products recover fusion coefficients exactly. The transport phase between homogeneous fibers is derived from the same three module identities and recorded with a separate derivation in the supplement. An untwisted convolution would be incorrect here.

6.2 The model of dimension 120

For $G = S_5$, $H = S_3 \times C_2$, let ε be parity and α the supplied normalized $\mathbb{F}_2$ two-cocycle, defined by a fixed section of the Schur double cover. Put

$$\omega(g,h,k) = (-1)^{\varepsilon(g)\alpha(h,k)}.$$

The explicit unit-modulus cochain ψ trivializes its restriction to H. The independent replay reconstructs S_5 from permutations and checks all 120^3 two-cocycle equations and all 12^3 algebra equations. The three stabilizer orders are $12, 2, 4$, with projective irreducible sizes $(1,1,1,1,2,2), (1,1), (2)$. The nondegenerate multiplier on the last V_4 gives $M_2(\mathbb{C})$. Simple dimensions are $1^4, 2^2, 6^3$, and the complete tensor is $R9_2_92$ under $(0,3,1,5,7,6,4,2,8)$.

6.3 Two models on the same group and subgroup

Take $G = S_4 \times C_2$, write e for the external bit and s for parity, and set

$$x = ((12),1), \quad y = ((34),0), \quad H = \langle x, y\rangle \simeq V_4.$$

Let X, Y be its two coordinates and pull the explicit Schur cocycle α back from S_4. The first associator and trivialization are

$$\omega_+ = (-1)^{e\smile(\alpha+e^2)}, \qquad \psi_+(a,b) = i^{X(a)Y(b)}.$$

They construct $R9_2_392$ under $(0,1,2,8,7,3,5,6,4)$.

For the second model let Ω be the copied eighth-root-valued S_4 cocycle from the lower-rank project. Its full table and original source hash are supplied; all 24^4 cocycle identities and every transported table entry are independently checked. Set

$$\omega_- = \Omega\,(-1)^{e\smile\alpha+s\smile e^2}.$$

In the order $(1,y,x,xy)$ of H, a trivialization $\psi_-(a,b)=\zeta_8^{q(a,b)}$ is given by

$$(q(a,b)) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 7 \\ 0 & 4 & 0 & 0 \\ 0 & 1 & 6 & 7 \end{pmatrix}.$$

The resulting tensor is $R9_2_1246$ under $(0,8,7,5,6,3,1,2,4)$. For each construction the double-coset sizes are $4,4,16,16,8$, with stabilizer orders $4,4,1,1,2$ and projective dimensions $(1,1,1,1),(2),(1),(1),(1,1)$. This gives type $1^4,2^3,4^2$. The independent checker verifies all 48^4 cocycle equations, the subgroup algebra and all tensor entries over the exact cyclotomic field. Neither construction relies on an unproved classification of dimension 24.

7 The half-E_6 family and the model inventory

7.1 Products and associative zesting

Let $\mathcal{E}$ be the unitary half-E_6 category and f its specified central fermion [6]. The products with Ising, $\mathrm{Rep}(S_3)$, $\mathrm{PSU}(2)_5$ and $\mathrm{Vec}(C_3)$ give respectively $R9_2_71, 79, 499, 2020$. Their tensors are direct Kronecker products. For $R9_2_1218$ use the Ising parity on $\mathcal{E}\boxtimes\mathrm{Ising}$ and zest the $(1,1)$ product by f. The associator cochain $\nu(a,b,c)=i^{abc}$ has coboundary $(-1)^{abcd}$ on parity bits, exactly canceling the central fermion crossing. All sixteen coherence equations are checked. The cochains and the initial categories are unitary.

7.2 A central-fermion shear with two coherent actions

On $\mathcal{D}=\mathcal{E}\boxtimes\mathrm{Vec}(C_2)$ define $T(Z,h)=(f^hZ,h)$. Its tensorator moves the inserted f^k left past Z using the inverse half-braiding, and multiplies the adjacent powers of f. The half-braiding identities prove its pentagon. Composition gives T^2 the identity tensorator multiplied by $(-1)^{hk}$, because $c_f=-1$. For either $u=i$ or $u=-i$, the components $\eta_{(Z,h)}=u^h$ are a monoidal isomorphism $T^2\to\mathrm{Id}$:

$$u^{h+k \bmod 2}(-1)^{hk}=u^hu^k.$$

The action equation $T\eta=\eta T$ holds because T preserves h. These are two coherent unitary C_2 actions.

The action fixes $(1,0),(f,0),(x,0),(x,1)$ and exchanges $(1,1),(f,1)$. Its equivariantization has four invertibles, four dimension-$(1+\sqrt{3})$ simples and one dimension-two object Y. Write ϵ for the acting sign and choose lifts a,b over x,xh. A normalized $t:fx\to x$ and $r=tc_x^{-1}:xf\to x$ satisfy

$$r(t\otimes 1_f)=-t(1_f\otimes r).$$

This is naturality and the half-braiding hexagon with $c_f=-1$; it makes f fix the two lifts of x and exchange the two lifts of xh. The rules include

$$a^2=1+f+2a,\quad Y^2=1+f+\epsilon+f\epsilon,$$

$$aY=b+\epsilon b,\quad bY=a+\epsilon a,\quad ab=Y+b+\epsilon b.$$

The sign of η changes the unit-pairing eigenvalue on the odd fixed simple. Thus one choice has $b^2 = 1 + f\epsilon + a + \epsilon a$, and the other $b^2 = \epsilon + f + a + \epsilon a$. These give $R9_2_96$ and $R9_2_1266$, including their distinct dualities. All coefficients, action coherence and forgetful identities are independently reconstructed.

7.3 All 32 positive tensors

Table 2 lists the accepted models. For group-theoretical and equivariantization entries, the archive supplies the actual cocycles, action data and projective stabilizer calculations, rather than an action on the Grothendieck ring alone. The exceptional AH3 entry is matched to its published categorical tensor. The WZW, subfactor, product and known near-group entries use their specified unitary realizations. The word "model" asserts existence for a based ring and makes no claim of novelty up to tensor equivalence.

Table 2: All unitary models. The final column is the source-to-model permutation; indices are zero-based.

Case	Construction	Basis map
R9_2_21	Parity-twisted V4 equivariantization of TY(C5)	013576428
R9_2_71	half-E6 x Ising	043276518
R9_2_79	half-E6 x Rep(S3)	043576218
R9_2_92	C(S5, sign cup Schur, S3 x C2, explicit trivialization)	031576428
R9_2_96	C2 equivariantization of the central-fermion shear on half-E6 x Vec(C2)	023567418
R9_2_369	Rep(PGL(2,7))	018245367
R9_2_389	Inversion orbifold of cyclic generalized Moore-Read category, dimension16	056241873
R9_2_392	C(S4 x C2, e cup (Schur + e^2), V4 graph subgroup, explicit trivialization)	012873564
R9_2_499	half-E6 x PSU(2)_5	035412678
R9_2_515	Diagonal-fermion quotient (Ising box PSU(2)_10)_1+psi box V10	054321687
R9_2_748	TY(C3)^S3 by inversion, parity coherence 0	016724583
R9_2_812	Rep(V4 semidirect Dih(C9))	016725384
R9_2_813	Rep(V4 semidirect Dih(C3 x C3))	016724583
R9_2_936	Rep(PSL(2,8))	085167243
R9_2_965	Davydov–Runkel rank-one extension of Gelaki A1, dim(A1)=20	023678541
R9_2_1016	C(S4,1,V4nonnormal,1)	067854123
R9_2_1211	TY(C3)^S3 by inversion, parity coherence 1	067245813
R9_2_1218	Fermionic associative C2 zest of half-E6 x Ising	025768143
R9_2_1246	C(S4 x C2, exotic S4 associator times e cup Schur plus sign cup e^2, V4 graph subgroup, explicit trivialization)	087563124
R9_2_1266	C2 equivariantization of the central-fermion shear on half-E6 x Vec(C2)	067832154
R9_2_1372	Asaeda-Haagerup category AH3	034128576
R9_2_1373	(Fib box Fib)^S3, sign quotient acts by swapping factors	045138762
R9_2_1374	D16 even category	087246135
R9_2_1396	Pulled-back LPR-24 action along Dih(C9)->S3	067253814
R9_2_1397	Pulled-back LPR-24 action along Dih(C3 x C3)->S3	067245813
R9_2_1557	C(GL(3,2),1,S4,1)	067124385
R9_2_1857	Spin(8)_2 WZW category equivariantized by order-three Dynkin triality	087213564
R9_2_1979	Izumi near-group (Q8,2)	013467582
R9_2_1983	Ordinary-swap equivariantization of the equal half-E6 diagonal-fermion product	013587642
R9_2_2020	half-E6 x Vec(C3)	045871236

Case	Construction	Basis map
R9_2_2029	Triality equivariantization of the Tannakian-zested Spin(8)_2 category	087352164
R9_2_2083	Rep(C19 semidirect C3, u=7)	012378564

8 What remains unresolved

Type	Suffixes of unresolved $R9_2_n$
Integral	12, 941, 959, 1965
Strictly weakly integral	585, 746, 1025, 1209, 1254, 1303, 1709, 1759, 1988
Other quadratic	137, 1607, 1848, 1975, 1976

For all eighteen rings general categorification and pseudo-unitary *existence* are equivalent by the indicated reduction. For the thirteen weakly integral rings a hypothetical category itself is pseudo-unitary. The five other quadratic reductions may use Galois conjugation; they do not assert positivity of every original category. General, pivotal, spherical, pseudo-unitary and unitary existence all remain undecided. Noncommutativity excludes braiding for 1848, 1965 and 1975. No braided model or exclusion is asserted for the other fifteen.

The integral cases 12 and 959 survive the bounded relative-cocycle work; 1965 survives the specified Morita and group-theoretical restrictions. The dimension-504 ring 941 is identified, in all 729 coefficients, with the ring posed in Question 1.14 of [12]. That identification does not settle its categorification. The nine strictly weakly integral rings retain the explicit extension or module-category gaps stated in their research records. Both tested mixed Moore–Read action systems fail only for the finite phase ansatzes used; this excludes no ambient fusion ring.

The remaining five quadratic rings pass the Galois restriction test. Bounded labelled-S searches on the three commutative ones produce no certified contradiction, with several time limits; this is not a proof of categorical feasibility. Correctly degree-weighted noncommutative constraints likewise leave 1848 and 1975 undecided. Failed model searches, feasible necessary matrices and unsuccessful separator optimizations do not change the accepted partition.

9 Exact evidence and reproduction

The public source is https://github.com/sebastienpalcoux/grothendieck-ring-classification/tree/main/evidence/rank9_m2_20261024. It exposes the accepted ledger, positive models, exact open tensors and readable proofs. Its checksummed replay archive retains every source cochain, certificate, constructor, independent audit and historical source context. Only `phase3/results/` is the current mathematical partition. Historical ledgers remain explicitly superseded; their unsupported half-E_6 claims are not used by the corrected verifier.

After unpacking the archive and installing its requirements, run

```
python3 phase3/verify_audit.py
python3 phase3/verify_audit.py -fresh
```

The first command checks all input identities, the partition, model permutations, explicit negative transfers and accepted receipts. The second reconstructs the corrected subring pass and the new constructors and independent exact checks. The archive manifest binds every member before fresh output is written. A replay is finite mathematical evidence and bookkeeping verification; it is not a formal proof-assistant development.

The fusion census repository publishes the same 2,225 supplied tensors through multiplicity two and the corresponding producer-run qualification. The public Atlas uses one rank/multiplicity

table and one case directory for all ranks. Every M2 case has its source identity, current decision, proof links and a downloadable Python tensor. The eighteen open cases retain their separate categorical-scope statements.

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